THE FIRST WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION
April 16, 1938

Morning Session

1. A solid is bounded by two bases in the horizontal planes $z = h/2$ and $z = -h/2$, and by such a surface that the area of every section in a horizontal plane is given by a formula of the sort

$$\text{Area} = a_0 z^3 + a_1 z^2 + a_2 z + a_3$$

(where as special cases some of the coefficients may be 0). Show that the volume is given by the formula

$$V = \frac{1}{6} h[B_1 + B_2 + 4M],$$

where B_1 and B_2 are the areas of the bases, and M is the area of the middle horizontal section. Show that the formulas for the volume of a cone and of a sphere can be included in this formula when $a_0 = 0$. (page 73)*

2. A can buoy is to be made of three pieces, namely, a cylinder and two equal cones, the altitude of each cone being equal to the altitude of the cylinder. For a given area of surface, what shape will have the greatest volume? (page 75)

3. If a particle moves in the plane, we may express its coordinates x and y as functions of the time t. If $x = t^3 - t$ and $y = t^4 + t$, show that the curve has a point of inflection at $t = 0$ and that the velocity of the moving particle has a maximum at $t = 0$. (page 76)

4. A lumberman wishes to cut down a tree whose trunk is cylindrical and whose material is uniform. He will cut a notch, the two sides of which will be planes intersecting at a dihedral angle θ along a horizontal line through the axis of the cylinder. If θ is given, show that the least volume of material is cut out when the plane bisecting the dihedral angle is horizontal. (page 77)

***The page number at the end of each problem indicates where the corresponding solution appears.**

5. Evaluate the following limits:

(i) $\lim_{n\to\infty} \frac{n^2}{e^n}$.

(ii) $\lim_{x\to 0} \frac{1}{x}\int_0^x (1 + \sin 2t)^{1/t}\, dt.$ (page 80)

6. A swimmer stands at one corner of a square swimming pool and wishes to reach the diagonally opposite corner. If w is his walking speed and s is his swimming speed ($s < w$), find his path for shortest time. [Consider two cases: (i) $w/s < \sqrt{2}$, and (ii) $w/s > \sqrt{2}$.] (page 81)

7. Take either (i) or (ii).

(i) Show that the gravitational attraction exerted by a thin homogeneous spherical shell at an external point is the same as if the material of the shell were concentrated at its center. (page 82)

(ii) Determine all the straight lines which lie upon the surface $z = xy$, and draw a figure to illustrate your result. (page 84)

Afternoon Session

8. Take either (i) or (ii).

(i) Let A_{ik} be the cofactor of a_{ik} in the determinant

$$d = \begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix}.$$

Let D be the corresponding determinant with a_{ik} replaced by A_{ik}. Prove $D = d^3$. (page 86)

(ii) Let $P(y) = Ay^2 + By + C$ be a quadratic polynomial in y. If the roots of the quadratic equation $P(y) - y = 0$ are a and b ($a \neq b$), show that a and b are roots of the biquadratic equation $P[P(y)] - y = 0$. Hence write down a quadratic equation which will give the other two roots, c and d, of the biquadratic. Apply this result to solving the following biquadratic equation:

$$(y^2 - 3y + 2)^2 - 3(y^2 - 3y + 2) + 2 - y = 0.$$

(page 87)

9. Find all the solutions of the equation

$$yy'' - 2(y')^2 = 0$$

which pass through the point $x = 1, y = 1$. (page 88)

10. A horizontal disc of diameter 3 inches is rotating at 4 revolutions per minute. A light is shining at a distant point in the plane of the disc. An insect is placed at the edge of the disc furthest from the light, facing the light. It at once starts crawling, and crawls so as always to face the light, at 1 inch per second. Set up the differential equation of motion, and find at what point the insect again reaches the edge of the disc. (page 90)

11. Given the parabola $y^2 = 2mx$, what is the length of the shortest chord that is normal to the curve at one end? (page 91)

12. From the center of a rectangular hyperbola a perpendicular is dropped upon a variable tangent. Find the locus of the foot of the perpendicular. Obtain the equation of the locus in polar coordinates, and sketch the curve. (page 92)

13. Find the shortest distance between the plane $Ax + By + Cz + 1 = 0$ and the ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$. (For brevity, let

$$h = 1/\sqrt{A^2 + B^2 + C^2} \quad \text{and} \quad m = \sqrt{a^2A^2 + b^2B^2 + c^2C^2}.)$$

State algebraically the condition that the plane shall lie outside the ellipsoid.

(page 93)